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YOLOv5x-based Brain Tumor Detection for Healthcare Applications

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Abstract

Brain tumors arise from the emergence of abnormal cells in brain tissue and are considered one of the most perilous conditions affecting individuals of all ages, including both children and adults. The disease advances swiftly, and the likelihood of survival diminishes significantly without prompt and adequate treatment. Hence, accurate diagnosis and meticulous treatment planning play a pivotal role in improving the patient's life expectancy. Neurologists and radiologists play a crucial role in the early detection of brain tumors. However, manually identifying and segmenting brain tumors from Magnetic Resonance Imaging (MRI) data poses significant challenges and is susceptible to inaccuracies. The need for an automated brain tumor detection method becomes imperative to achieve early detection of brain tumors.

Objective: The objective of the paper is to measure the capability of You Only Look Once version 5 (YOLOv5x) in brain tumor detection in the early stage so that patients can be treated accordingly.

Methods: YOLOv5x is examined; Brats (Brain Tumor Segmentation) image and roboflow dataset has been used. Model performance is evaluated using precision, recall rate, F1 score, and Mean Average Precision (mAP).

Results: YOLOv5x exhibited precision (98.7), recall (95.6), F1-score (93), mAP at a learning rate of 0.5 (98.4), and the total time taken for implementation of work is 193.20 minutes.

Conclusion: YOLOv5x showed improved performance for the detection of brain tumors on dynamic contrast-enhanced MRI when compared with state of art existing work. It is also the fastest and accurate method indicating a greater potential for clinical application.

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Keywords: Brain Tumor, Brain Tumor Detection, You Only Look Once (YOLOv5x), Deep Learning.

1. Introduction

A brain tumor is an irregular development of cells in the brain or the central nervous system. The tumors could be cancerous or noncancerous. It can develop in different parts of the brain and can have varying symptoms depending on their location and size. The indicators of brain tumors include headache, annexations, vision and hearing problems, changes in personality, difficulty in coordination, or numbness in part of the body [1]. Several factors can cause brain tumors such as family history, contact with radiation, genetic conditions, immune system disorder, environmental toxins, age of the patient, and many more. Treatment options for brain tumors include surgery, radiation therapy, chemotherapy, and targeted drug therapy. After the detection of the tumor proper treatment depends on many factors like the type of tumor, size of the tumor, the position of the tumor, the health of

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the patient, and many more [2]. This cancer is a relatively rare type of cancer, but it can be serious and often result in deadly conditions. Here are some statistics on brain tumors worldwide [3][8]:

- The incidence of brain tumors varies by country, with higher rates reported in developed countries.
- It could occur in any age, but most cases are found in older age group people. The median age at diagnosis is 60 years.
- The five-year survival rate for all types of brain cancer is around 36%. However, survival rates can vary widely depending on the type, size, location, stage of cancer, and health of the patient.
- Meningiomas are more common in women and acoustic neuromas are more common in Caucasians.
- Brain tumors can be difficult to treat, as they are often located in delicate and important areas of the brain.

Premature exposure to brain cancer is necessary for efficacious treatment. Deep Learning (DL) models have shown great potential in the detection of various medical conditions, including brain tumors. In this research work, the authors used the DL model using YOLOv5x architecture for the recognition of brain cancer. The architecture of the YOLOv5x has been modified and data augmentation is also applied to improve the accuracy and detection rate. The research work is evaluated and compared with existing methods for its validation.

The paper is structured into 6 sections. Section 1 introduces the brain tumor and the YOLOv5x deep learning model. Section 2 comprises a literature review covering the latest developments in deep learning techniques. Section 3 presents the problem statement. In Section 4, the preliminary aspects of the proposed research work are discussed. Section 5 explores result analysis, presenting performance metrics and comparing them with state-of-the-art work. The final section provides a conclusion and discusses the future scope of the research.

2. Literature Review

The use of AI in the field of radiology has significantly improved the diagnosis of Brain Tumors (BTs). Early detection of diseases can lead to better treatment outcomes, and AI systems have proven to be time and cost-effective tools in healthcare. However, traditional methods of tumor detection using MR imaging have not compensated for the problems associated with different MR imaging types. To overcome these issues, the DL method has been used in this study to develop an improved YOLOv5-based BT detection system using MR images. The Hybrid Grid Search Optimizer Algorithm (HGSOA) has been used to optimize the outcome of the proposed work by tuning its hyperparameters. The outcome of the research work has been evaluated by localizing images for tumor region segmentation using McCulloch's algorithm [4]. Brain tumors are a serious medical condition, caused by mutations in brain cells leading to uncontrolled cell division. In recent years, computer technology and machine learning algorithms have an important role in the diagnosis of brain cancer. Methodology: We utilized the YOLOv5 architecture for developing the brain tumor detection model. The open dataset of tumorous images with corresponding masks and MRI dataset were used for model training. The study compared the performance of several YOLOv5 models, including YOLOv5m, YOLOv5l, and YOLOv5x, to localize brain tumors. The performance metrics were evaluated, and the inference time was also calculated. YOLOv5x performed better than the three other models of the YOLOv5 [5].

Brain tumors are caused by the expansion of aberrant brain cells. Premature exposure to brain cancer is important, but manual recognition and classification from MRI data take time and lead to errors. To address this issue, automated brain tumor detection systems have been developed. Instance segmentation is a computer vision technique that assigns different IDs to objects in an image, even if they belong to the same class. This technique has attracted a lot of interest in current times. In the work, brain tumor segmentation is performed using the YOLO algorithm. YOLOv5, the latest version of the YOLO algorithm, is utilized to develop a brain tumor detection system. The YOLOv5 algorithm works on large datasets and text files also. The model developed in this study has three classes: Meningioma, Pituitary, and Glioma. The simulation of work proved that the proposed model achieved high accuracy in tumor detection [6]. Early detection of breast tumors is crucial to improving survival rates. Mammography is a proven screening technique, but accurately detecting and classifying tumors in mammograms remains challenging. Previous studies have struggled with high FP and FN rates, also the Matthews Correlation Coefficient (MCC) was reduced. To address these limitations, this study employs a modified YOLOv5 for tumor identification. The CBIS-DDSM

dataset was used, and a variety of preprocessing techniques were applied, including image enhancement and muscle and label removal. The proposed model is compared to YOLOv3 and faster RCNN, and the results demonstrate superior performance, with a 96% mAP, and 93.50% MCC [7].

Breast tumor is a widespread disease and the prominent reason for demise among women worldwide. With limitations of the current imaging approach, researchers have been exploring alternative technologies for early detection of tumors. This study proposes a novel technique that employs non-ionizing microwave signals and deep learning algorithms for detecting and classifying breast tumors at an early stage. The proposed method involves using backscatter signals obtained from antenna simulation in CST microwave studio software and processing them through MATLAB programming environment to create an image. The concentration in the image indicates the exact location of the tumor. The YOLOv5 deep learning model was then utilized to automatically classify the tumor category from the microwave images. For training, 50 image samples were preprocessed and augmented to generate a sample of 1000 images. The promising results of this study suggest that non-ionizing microwave signals and deep learning algorithms have the prospective for early detection and classification of breast cancer [9].

3. Problem Statement

Brain cancer is the leading reasons of disease worldwide. Premature recognition and accurate diagnosis of brain cancer are important for effective diagnosis and result in better patient outcomes. Currently, radiologists and other medical professionals rely on a combination of visual inspection and various imaging modalities to identify brain tumors. However, these processes could take more time and sometimes may result in false detection. Deep learning has shown great potential in automating the process of brain cancer recognition. In the work, YOLOv5x deep learning model has been used to identify brain cancer from medical imaging data.

4. Preliminary of Proposed Research Work

YOLOv5x is a DL object recognition model that is built upon the YOLO family of models. YOLOv5x is the largest and most powerful of the YOLOv5 models, featuring over 170 million parameters. The architecture of YOLOv5x follows the general YOLOv5 structure, which consists of a backbone network, a neck network, and a head network represented in Figure 1. The backbone network is responsible for retrieving features from the input image, while the neck network is utilized to fuse features from diverse levels of the backbone network. Finally, the head network generates bounding box predictions and class probabilities for each detected object [10].

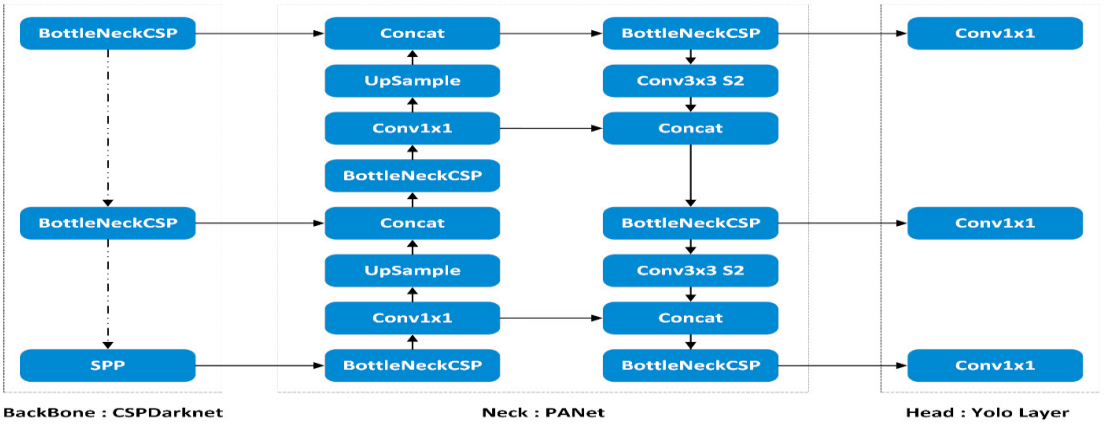


Fig. 1. Layered Architecture of YOLOv5x

The backbone network of YOLOv5x is based on Cross-Stage Partial (CSP) architecture, which is designed to improve the efficiency of deep neural networks. CSP consists of two pathways that process the input image in parallel, which allows for better feature extraction while reducing computation time. The neck network of YOLOv5x is built on top of the backbone network and consists of three attention CSP modules. These modules are used to fuse features from diverse levels of the

backbone network, permitting the system to identify objects at dissimilar scales. Finally, the head network of YOLOv5x consists of a set of fully connected layers that generate the final bounding box predictions and class probabilities. YOLOv5x uses anchor-based detection, which means that it generates predictions based on predefined anchor boxes that are optimized during training [12]. Figure 2 represents the proposed flowchart for the work.

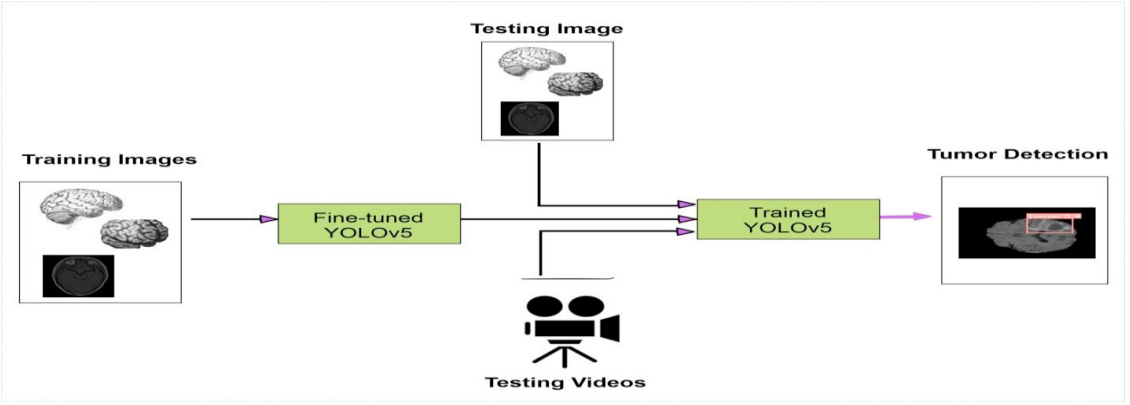


Fig. 2. Proposed Flowchart

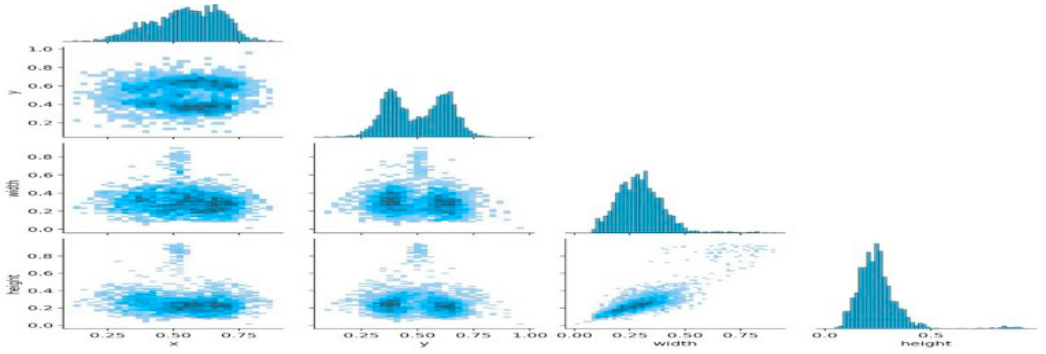


Fig. 3. Distribution of the Sizes and Positions of Brain Tumor in Dataset

The predictions made by the head network of YOLOv5x are post-processed to enhance the final results by eliminating redundant detections. This is accomplished using a Non-Maximum Suppression (NMS) algorithm, which removes overlapping detections and retains the most confident detections for each object. YOLOv5x employs a layered architecture that empowers it to achieve top-notch performance in object detection tasks, exhibiting high precision and efficiency [13]. Figure 3 represents the distribution of size and position of brain cancer in the dataset.

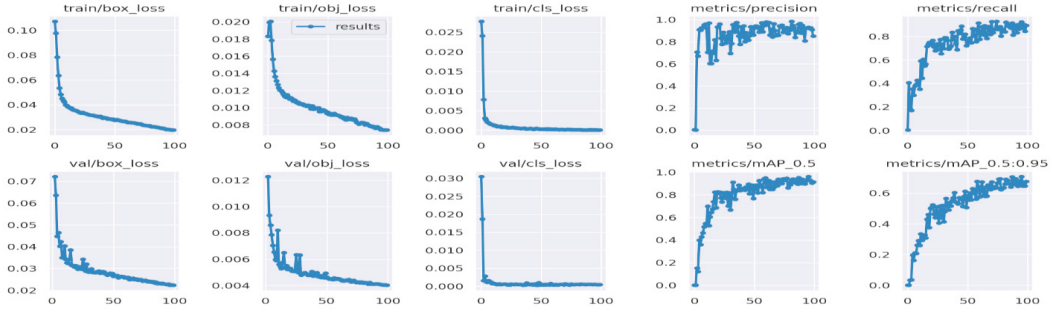


Fig. 4. Training and Validation Loss

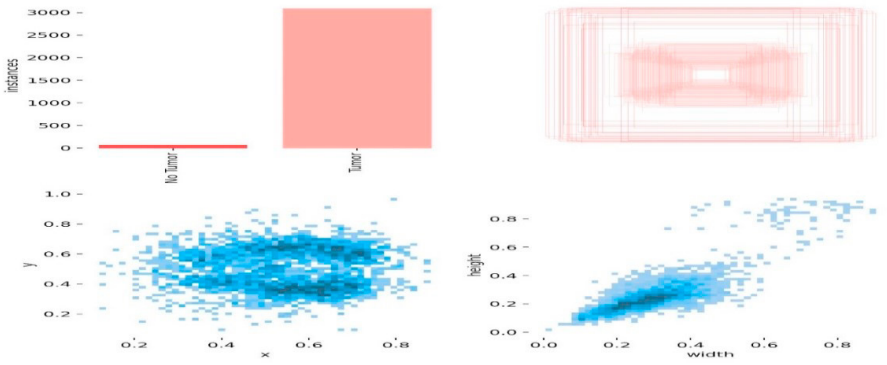


Fig. 5. Distribution and Histogram of Brain Tumor in Training Set

Box loss, as shown in Figure 4, measures the change between the expected and true bounding box coordinates of an object. Fig. 5 represents the distribution and histogram of the brain tumor during the training set. It is typically calculated using a regression loss function such as MSE or plane L1 loss. Equations 1 and 2 represent the bounding box and object box mathematically.

$$L_{box} = \lambda_{coordinate} \sum_{n=0}^{k^2} \times \sum_{m=0}^{l} [\, I_{n,m}^{obj} [(o_n - \hat{o}_n)^2 + (h_n - \hat{h}_n)^2]] + \lambda_{coordinate} \sum_{n=0}^{k^2} \times \sum_{m=0}^{l} [\, I_{n,m}^{obj} [(\sqrt{c_n} - \sqrt{\hat{c}_n})^2 + (\sqrt{e_n} - \sqrt{\hat{e}_n})^2]] \quad (1)$$

where k is the numeral of grid cells in the input, I am the numeral of bounding boxes expected by individual grid cells, $\hat{o}$, $\hat{h}$, $\hat{c}$, $\hat{e}$ are the predicted coordinates of the center, width, and height of the box in cell n, and o_n, h_n, c_n, e_n are the ground-truth coordinates of the center, width, and height of the box in cell n. The term $I_{n,m}^{obj}$ is a pointer function that equals 1 if the m bounding box in cell n is accountable for detecting an object or 0 otherwise. The parameter $\lambda_{coordinate}$ controls the weight of the box localization loss.

$$L_{obj} = \sum_{n=0}^{k^2} \times \sum_{m=0}^{l} 1_{n,m}^{obj} \left[\log\left(\sigma(\hat{\imath}_{n,m})\right) + \left(1 - 1_{n,m}^{obj}\right) \log\left(1 - \sigma(\hat{\imath}_{n,m})\right) \right] \quad (2)$$

Where $\hat{\imath}_{n,m}$ is the predicted object score for the mth bounding box in cell n, and σ is the sigmoid function. In the term $1 - 1_{n,m}^{obj}$ is a pointer function that equals 1 if the mth bounding box in cell n is not accountable for detecting an object and 0 otherwise.

$$loss_{YOLOv5} = loss_{bounding\ box} + loss_{classification} + loss_{confidence} \quad (3)$$

$$loss_{\text{bounding box}} = \lambda_{if} \sum_{a=0}^{b^2} \times \sum_{c=0}^{d} E_{a,c}^{g} h_g (2 - K_a X n_a) \left[(x_a - \acute{x}_a^c)^2 + (y_a - \acute{y}_a^c)^2 + (w_a - \acute{w}_a^c)^2 + (h_a - \acute{h}_a^c)^2 \right] \quad (4)$$

$$loss_{classification} = \lambda_{classification} \sum_{a=0}^{b^2} \times \sum_{c=0}^{d} E_{a,c}^{g} \times \sum_{C \in c_l} L_a(c) \log\left(LL_a(c)\right) \quad (5)$$

$$loss_{confidence} = \lambda_{confidence} \sum_{a=0}^{b^2} \times \sum_{c=0}^{d} E_{a,c}^{confidence} \ (c_i - c_l)^2 + \lambda_g \sum_{a=0}^{b^2} \times \sum_{c=0}^{d} E_{a,c}^{g} \ (c_i - c_l)^2 \quad (6)$$

Loss function in YOLOv5x has a significant role in enhancing the outcomes. It is the combination of boundary box, classification, and confidence as represented by Equation 3. Equation 4 shows how the bounding box can be calculated. Where h′ and w′ show the height and width of the target object in the input image. x_a and y_a represent the coordinates of the input image. λ_{if} indicates whether the target object exists in the image or not. $\lambda_{confidence}$ in Equations 5 and 6 represents the loss coefficient. $\lambda_{classification}$ is the classification loss coefficient, c_l is class, and c is the confidence score.

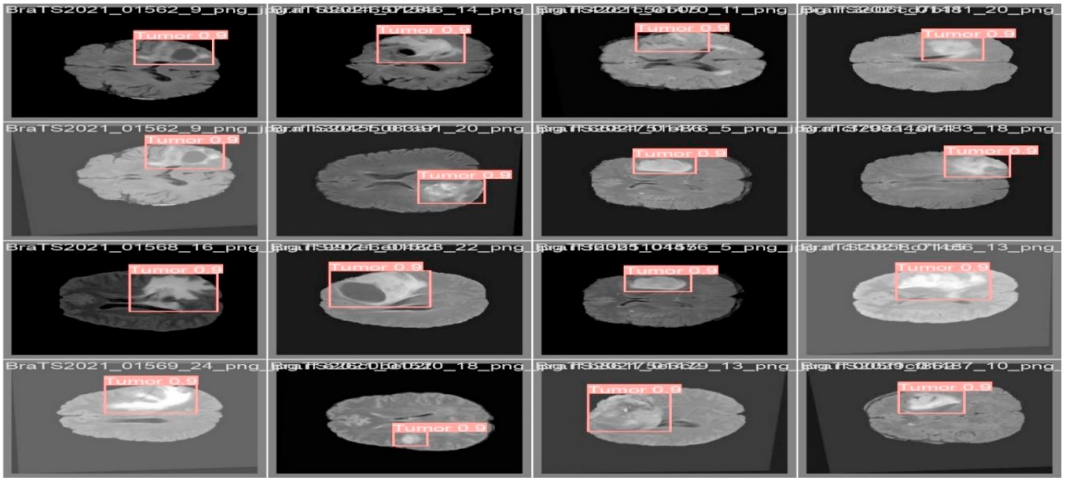


Fig. 6. Brain Cancer Detection

Object loss is the loss function that is used to decide if an object exists or not in the input image. It is calculated using a sigmoid activation function and binary cross-entropy loss. The purpose of object loss is to adjust the model's parameters so that it can more accurately identify the occurrence or nonexistence of an object in an image. Class loss is used for multi-class classification responsibilities. It measures the difference between the expected probabilities and the true probabilities of an object fitting to different classes. The class loss is computed using a softmax activation function and categorical cross-entropy loss. The objective of class loss is to optimize the model's parameters so that it can better classify objects into their respective categories. For the reduction of object loss in the proposed work we have used high quality dataset and regress training process.

As shown in Figure 6, the results of YOLOv5x detection demonstrate that the deep learning algorithm can detect drones with high accuracy. YOLOv5x can achieve top-level precision on various benchmark datasets. YOLOv5x uses CSP architecture, which is designed to enhance the efficiency of Deep Neural Networks (DNN). This allows YOLOv5x to maintain fast inference times even with lots of parameters. It has over 170 million parameters, making it the largest and most powerful of the YOLOv5 models. It uses anchor-based detection, which allows it to generate predictions based on predefined anchor boxes that are optimized during training. By using data augmentation the accuracy and detection rate of the YOLOv5x can be improved. It can detect objects at different scales by fusing structures from dissimilar levels of the backbone network using cross-stage partial attention (CSP) modules. It can perform object detection in real-time on a variety of devices, including GPUs and CPUs. It can handle input images of various sizes and aspect ratios, making it suitable for applications.

5. Result Analysis and Discussion

Precision: BRATs and the Roboflow brain dataset of annotated brain MRI images have been used for the computation of performance metrics. BRATs dataset contains images of brain tumors for detection and segmentation. It also contains

surrounding brain tissues along with brain tumor images. It contains 4 MRI modalities and different types of tumors. It also provides annotation to the images. Roboflow is used to manage and pre-process images provided through the BRATs dataset.

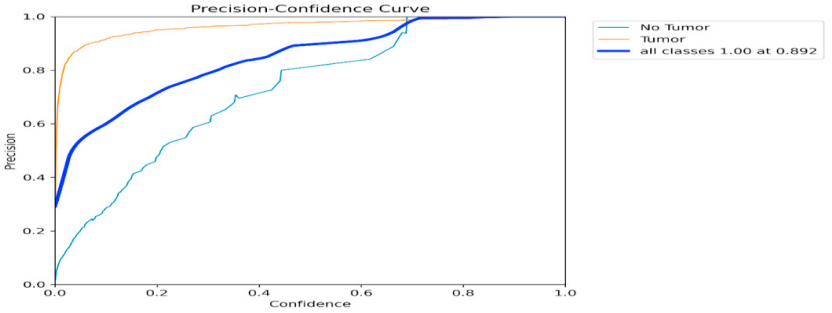


Fig. 7. Precision Graph

The dataset is split into training and testing groups. Yolov5x is skilled in the training set using annotated images. The model is run on the testing set and evaluated its performance using ground truth annotations. The number of True Positives (TP) and False Positives (FP) are calculated for each image. A TP means it detected the tumor correctly, and FP means incorrect detects of the tumor. Calculate an overall number of TP and FP across images in the testing set for computing precision [14]. The precision obtained for the work is 98.7% as shown in Figure 7.

Recall: It is another important metric for evaluating the presentation of brain cancer recognition using Yolov5x. Recall is the ratio of TP to the full number of genuine positives in the ground accuracy annotations. In the context of brain tumor detection, recall represents the system's capability to detect all of the tumors in the images.

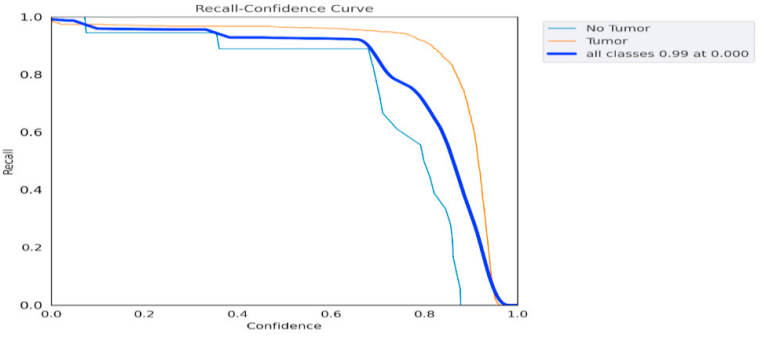


Fig. 8. Recall Graph

Yolov5x has been trained on the training set using annotated images. The trained model is run on the testing set and its performance is evaluated using the ground truth annotations. The number of TP, FP, True Negatives (TN), and False Negatives (FN) are calculated for each image [15]. The recall for the work is calculated which is 95.6% as shown in Figure 8 as well.

F1 Score: It is a commonly used evaluation metric in deep learning for measuring precision and recall. The F1-score of YOLOv5x will depend on the specific dataset and task it is being evaluated on.

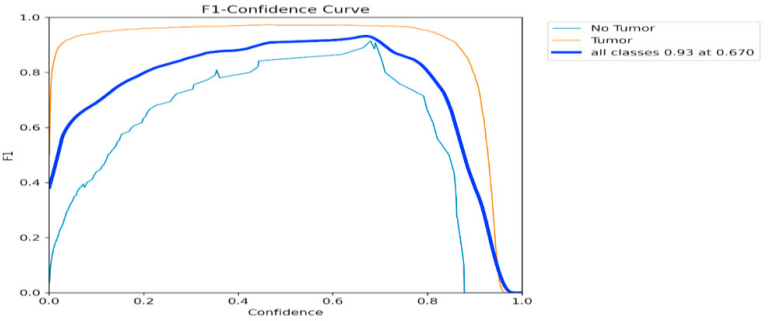


Fig. 9. F1-Score Graph

To calculate the F1 score for YOLOv5x, the model needs to train on the dataset, and then evaluate its performance on the test set. The F1 score is deliberate as the harmonic mean of precision and recall. Figure 9 represents the F1-score and it is calculated 93.00%.

mAP: It measures the average precision over a range of different confidence thresholds. It splits the dataset into training and testing sets. The Yolov5x model has been trained on the training set using annotated images. Run the trained model on the testing set and obtain the model's confidence scores for each detection. Detections are sorted by confidence scores in descending order.

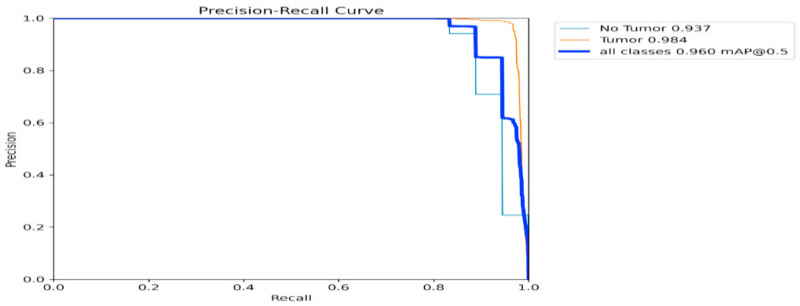


Fig. 10. mAP Graph

The mAP is calculated by taking the mean of the AP scores across all classes. The final mAP score represents the model's overall performance for brain tumor detection. The calculation of mAP requires a threshold value for deciding if detection is considered TP or FP. Figure 10 represents mAP at a learning rate of 0.5 which is found to be 98.4%. The total time taken to implement the work is 3 hours 13 minutes and 20 seconds.

Comparison: The article proposes YOLOv5x, an advanced object detection algorithm. The proposed work is skilled on a dataset of brain MRI images and achieves a high accuracy in detecting brain tumors. Compared to existing works in the field of brain cancer identification, the proposed model offers several advantages. First, it uses the object detection method, which has been shown to achieve high accuracy in various computer vision tasks.

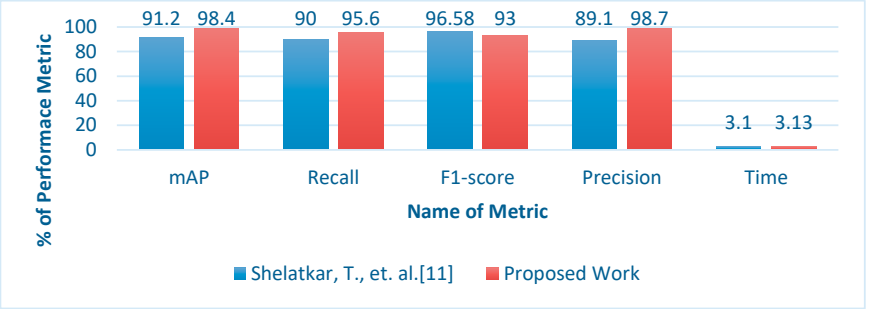


Fig. 11. Comparison Graph

Second, it utilizes transfer learning, which allows the model to influence pre-trained weights from other tasks to improve its performance on the specific task of brain tumor detection. Figure 11 illustrates a quantitative analysis, showcasing the presentation of the proposed method in contrast to Shelatkar, T., et al.'s work [11]. The evaluation is based on metrics such as mAP at a learning rate of 0.5, recall, F1-Score, precision, and time in hours. The proposed approach demonstrates superior results, with a 7.2% higher mAP compared to the findings in [11]. Additionally, the recall rate surpasses the referenced work by 5.6%, and the F1-Score also outperforms the existing work. The precision metric indicates a notable improvement of 9.6%. Although the proposed model exhibits commendable performance across these metrics, it is important to note a slightly higher time requirement—specifically, an additional 3 minutes compared to the existing work. Despite this, the proposed model is characterized as relatively lightweight, making it well-suited for deployment on devices with resource constraints.

Research Outcomes:
Increased Accuracy: As a result of optimization of model architecture, training data, and hyperparameters the accuracy of detection of brain tumors improved. For improvement of accuracy and detection rate, we fine-tuned the architecture of YOLOv5x along with data augmentation.
Validation on Larger Datasets: The YOLOv5x model is tested on larger and more diverse datasets to determine its performance in real environment settings and to confirm generalizability across different patient populations.
Comparison with other DL Models: To measure the effectiveness of research work it is equated to existing state of art work. The research work is found to be more effective in terms of accuracy, time, and memory
Clinical Validation: When YOLOv5x is compared with radiologists or pathologists and its clinical utility in aiding diagnosis and treatment decisions then YOLOv5x justifies and validates its use.
Integration with Other Diagnostic Tools: YOLOv5x has the potential to integrate with other diagnostic tools, such as radiomics or genomics, to enhance the accuracy and reliability of brain cancer identification.

6. Conclusion

In conclusion, the detection of brain tumors is a crucial task in medical imaging that can aid in early diagnosis and treatment planning. The use of deep learning models, such as YOLOv5x, has shown promising results in the accurate detection of brain tumors in medical images. By training the YOLOv5x on a big dataset of brain MRI images, it can effectively recognize the position and size of the cancer in the image with high accuracy. This could decrease the time and effort requisite for manual understanding of medical images, and provide faster and more accurate diagnoses. Moreover, the utilization of DL models like YOLOv5x in the medical field can help to overcome the restrictions of traditional methods and enhance the overall eminence of healthcare. For the proposed work precision was calculated as 98.7%, Recall as 95.6%, F1-Score as 93%, and mAP at learning rate 0.5 is 98.4%. As these models continue to evolve and improve, they have the potential to become a vital tool for medical professionals in the diagnosis and treatment of numerous medical circumstances, including brain tumors.

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